ANNASÈTU ([redacted]) • SIH 2026 Finalist (SIH26032)

Member #1 Profile Dossier

Team Lead & Full-Stack System Architect

Focus: System Architecture, Real-Time WebSockets & Core Full-Stack Pipeline

Executive Summary & Core Responsibility:
Architected the end-to-end 5-stage async state machine, WebSocket synchronization engine, and unified application lifecycle.

1. Technology Stack & Architectural Components

Domain / Layer	Specific Technologies, Libraries & Frameworks Used
Backend Architecture	FastAPI (Python 3.11), Uvicorn ASGI Server, AsyncIO event loop
Real-Time Protocols	WebSockets (Native Bi-directional Pub/Sub Manager with auto-reconnect)
Database & State	SQLite3 with ACID transactions, Row Factory, Foreign Keys & WAL mode
API Design	RESTful APIs, OpenAPI 3.0 / Swagger, CORS Middleware, Structured JSON contracts
Integration & Pitch	End-to-End full stack orchestration, system integration testing, pitch delivery

2. Key Modules Developed & Production Deliverables

- **FastAPI Core Architecture (main.py):** Designed modular routing, asynchronous lifespan startup handlers, static asset mounting, and Jinja2 template rendering.
- **Real-Time WebSocket Engine (websocket_manager.py):** Developed connection manager supporting global broadcast channels, targeted token-subscriber queues, and center-specific telemetry streams.
- **5-Stage Procurement State Machine:** Implemented automated lifecycle tracking (REGISTERED -> GATE_ENTRY -> WEIGHBRIDGE -> QUALITY_CHECK -> PAYMENT_PROCESSED) ensuring tamper-proof state transitions.
- **Full-Stack Integration & Coordination:** Coordinated frontend-backend interface contracts, resolved cross-origin and real-time state synchronization across all modules.

3. Direct Codebase Ownership & Modified Files

Source Files:
`backend/app/main.py` • `backend/app/websocket_manager.py` • `backend/app/models.py` • `run.py`

4. SIH Jury Defense & Live Evaluation Points (What to Speak Tomorrow)

- **Presentation Pitch Point:** Explain how the async WebSocket engine achieves sub-50ms live updates without database lock contention.
- **Presentation Pitch Point:** Describe the architecture choices (FastAPI + ASGI over traditional synchronous frameworks) to handle thousands of concurrent mandi requests.
- **Presentation Pitch Point:** Demonstrate how state transitions enforce atomic database commits and instant global broadcasts.